

Lecture 5: Superconducting Circuits

Deep Derivations — From Circuit Quantization to the Transmon

Quantum Computing with Superconducting Qubits: Basic Concepts

Prof. Stefan Filipp — Walther-Meißner-Institut / TUM

Comprehensive Study Guide — Part II (Circuit Quantization Onward)

Contents

1	Recap: The Simple LC Oscillator (Foundation)	3
1.1	The Key Identifications	3
2	Circuit Quantization Recipe — Full Treatment	4
2.1	The Problem Statement	4
2.2	Graph-Theoretic Setup	4
2.3	Spanning Tree	4
2.4	Node Fluxes: The Generalized Coordinates	5
2.5	Branch Fluxes in Terms of Node Fluxes	5
2.6	Step-by-Step: From Circuit to Hamiltonian	5
3	Worked Example: Two Nodes Coupled by a Capacitor	6
3.1	Step 1–4: Setup	6
3.2	Step 5–6: Energies in Terms of Node Fluxes	6
3.3	Step 7: Lagrangian	7
3.4	Step 8: Conjugate Momenta (Node Charges)	7
3.5	Step 9: Legendre Transform — The Hamiltonian	7
3.6	Step 10: Quantization	8
4	The Quantum Harmonic Oscillator — Full Derivation	9
4.1	From Hamiltonian to Ladder Operators	9
5	Physical Realizations and the Qubit Problem	11
5.1	Lumped-Element and Transmission-Line Resonators	11
5.2	Why a Harmonic Oscillator Cannot Be a Qubit	11
6	Phase and Flux Quantization	12
6.1	Flux Quantization in a Superconducting Loop	12
6.2	Phase-Number Commutator	12
7	The Josephson Effect — Full Derivation	13
7.1	Physical Setup	13
7.2	The Hilbert Space of the Junction	13
7.3	The Josephson Hamiltonian (Tunneling Hamiltonian)	13
7.4	Phase States and the Cosine Potential	13
7.5	The First Josephson Relation — Full Derivation	15
7.6	The Second Josephson Relation — Full Derivation	15

8	The Josephson Inductance — Nonlinear Circuit Element	17
8.1	Derivation from the Josephson Relations	17
8.2	Physical Interpretation	17
8.3	From Quadratic to Cosine Potential	17
9	The Full Junction Hamiltonian and the Transmon	19
9.1	The Junction as a Parallel L_J – C Circuit	19
9.2	The Two Competing Energy Scales	19
9.3	Limiting Regimes	19
9.4	Transmon Spectrum — Deriving ω_q and α	20
10	Synthesis: The Complete Logical Chain	22
10.1	What Makes This Work	22
10.2	Key Numbers for a Typical Transmon	22
11	Complete Formula Reference	23
11.1	Fundamental Constants	23
11.2	Harmonic Oscillator (LC Circuit)	23
11.3	Josephson Junction	23
11.4	Transmon Qubit	23

1. Recap: The Simple LC Oscillator (Foundation)

Before diving into the general recipe, let us crystallize the logic of the simple LC circuit so that the generalization feels natural.

1.1 The Key Identifications

For a parallel LC oscillator (one active node above ground), we chose the **node flux** Φ as the generalized coordinate. The logic:

- (i) **Faraday's law** gives $V = \dot{\Phi}$, so flux is the time-integral of voltage.
- (ii) The **capacitive energy** $T = \frac{1}{2}C\dot{\Phi}^2$ depends on $\dot{\Phi}$ (velocity-like) $\Rightarrow$ this is *kinetic energy*, and C plays the role of *mass*.
- (iii) The **inductive energy** $W = \frac{\Phi^2}{2L}$ depends on Φ itself (position-like) $\Rightarrow$ this is *potential energy*, and $1/L$ plays the role of the *spring constant*.
- (iv) The **Lagrangian** is $\mathcal{L} = T - W = \frac{1}{2}C\dot{\Phi}^2 - \frac{\Phi^2}{2L}$.
- (v) The **conjugate momentum** is $Q = \frac{\partial \mathcal{L}}{\partial \dot{\Phi}} = C\dot{\Phi} = CV$ — the charge on the capacitor.
- (vi) The **Hamiltonian** via Legendre transform: $H = Q\dot{\Phi} - \mathcal{L} = \frac{\Phi^2}{2L} + \frac{Q^2}{2C}$.
- (vii) **Quantization**: promote $[\hat{\Phi}, \hat{Q}] = -i\hbar$, yielding $\hat{H} = \hbar\omega(\hat{a}^\dagger\hat{a} + \frac{1}{2})$ with $\omega = 1/\sqrt{LC}$.

This procedure generalizes directly to arbitrary networks. The only complication is bookkeeping: with N active nodes, we get N coupled degrees of freedom, and the capacitance matrix couples them.

2. Circuit Quantization Recipe — Full Treatment

2.1 The Problem Statement

We are given an arbitrary lumped-element circuit composed of capacitors, inductors, and Josephson junctions, all made from superconducting materials. Our task: systematically derive the quantum Hamiltonian $\hat{H}(\hat{\phi}_1, \hat{Q}_1, \dots, \hat{\phi}_N, \hat{Q}_N)$.

References: Devoret, *Quantum Fluctuations in Electrical Circuits*, Les Houches (1995); Vool & Devoret, arXiv:1610.03438 (2016); Burkard, Koch & DiVincenzo, PRB 69 (2004); Lev Bishop thesis, Yale (2010).

2.2 Graph-Theoretic Setup

Model the circuit as a **graph**:

- **Nodes:** junction points where wires meet. Label them $0, 1, 2, \dots, N$, where node 0 is chosen as ground.
- **Branches:** two-terminal circuit elements connecting pairs of nodes. Each branch b has:
 - A voltage $v_b(t)$ across it (from its “+” to “−” terminal),
 - A current $i_b(t)$ through it.

It is more natural to work with the time-integrated quantities:

$$\Phi_b(t) = \int_{-\infty}^t v_b(t') dt' \quad (\text{branch flux}), \quad (1)$$

$$Q_b(t) = \int_{-\infty}^t i_b(t') dt' \quad (\text{branch charge}). \quad (2)$$

Every element is either **capacitive** (its constitutive relation is $v_b = f(Q_b)$, e.g. $v_b = Q_b/C$) or **inductive** (its constitutive relation is $i_b = g(\Phi_b)$, e.g. $i_b = \Phi_b/L$ for a linear inductor, or $i_b = I_c \sin(\Phi_b/\varphi_0)$ for a Josephson junction).

2.3 Spanning Tree

A **spanning tree** $\mathcal{T}$ is a subgraph that:

1. Contains *all* $N + 1$ nodes,
2. Is *connected* (there is a path between any two nodes),
3. Contains *no loops*.

A tree with $N + 1$ nodes has exactly N branches. Branches in $\mathcal{T}$ are called **tree branches**; the remaining branches are **closure branches** (or *chords*). Adding any single closure branch to $\mathcal{T}$ creates exactly one loop — this is called a **fundamental loop**.

The choice of spanning tree is a *gauge choice* — different trees lead to different (but physically equivalent) representations of the Hamiltonian, related by canonical transformations. In practice, one typically chooses a tree that:

- Includes as many capacitors as possible (so that the “kinetic energy” matrix is as simple as possible),
- Places Josephson junctions as closure branches when external flux bias is present, to

keep track of the flux constraint.

2.4 Node Fluxes: The Generalized Coordinates

For each active node $n \in \{1, 2, \dots, N\}$ (all nodes except ground), define the **node flux**:

$$\phi_n(t) = \sum_{b \in \mathcal{P}(0 \rightarrow n)} S_{nb} \Phi_b = \sum_{b \in \mathcal{P}(0 \rightarrow n)} S_{nb} \int_{-\infty}^t v_b(t') dt' \quad (3)$$

where $\mathcal{P}(0 \rightarrow n)$ is the *unique path in the tree $\mathcal{T}$* from ground (node 0) to node n , and $S_{nb} = \pm 1$ depending on whether the path traverses branch b in the “+” or “−” direction. If the path does not pass through b , then $S_{nb} = 0$.

Physically, ϕ_n is the time-integral of the voltage from ground to node n along the tree path. The node fluxes $\{\phi_1, \dots, \phi_N\}$ form a complete set of independent coordinates for the circuit.

2.5 Branch Fluxes in Terms of Node Fluxes

For any branch b connecting nodes n and n' :

$$\Phi_b = \phi_n - \phi_{n'} + \Phi_{l(b)}^{\text{ext}} \quad (4)$$

where:

- If $b \in \mathcal{T}$ (a tree branch): $\Phi_{l(b)}^{\text{ext}} = 0$.
- If $b \notin \mathcal{T}$ (a closure branch): $\Phi_{l(b)}^{\text{ext}}$ is the externally applied magnetic flux threading the fundamental loop $l(b)$ created by adding b to $\mathcal{T}$.

This is simply Kirchhoff’s voltage law (KVL) applied around each fundamental loop: the sum of branch fluxes around a loop equals the external flux through that loop.

2.6 Step-by-Step: From Circuit to Hamiltonian

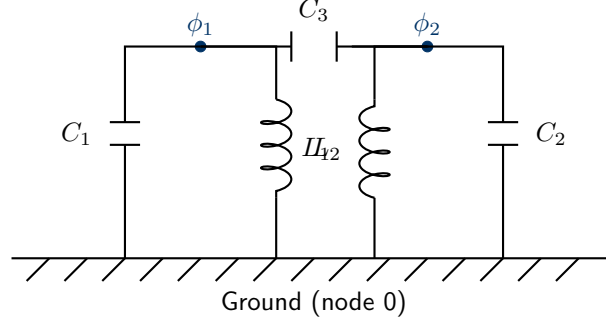
The Complete Recipe

- Step 1: Draw** the circuit as a network of two-terminal elements.
- Step 2: Choose ground** (node 0). All other nodes are active.
- Step 3: Choose a spanning tree $\mathcal{T}$.**
- Step 4: Define node fluxes ϕ_n** for each active node.
- Step 5: Compute energies:** kinetic energy T (capacitive) in terms of branch voltages $\dot{\Phi}_b$, potential energy W (inductive) in terms of branch fluxes Φ_b .
- Step 6: Express T and W** in terms of $\{\phi_n, \dot{\phi}_n\}$.
- Step 7: Form Lagrangian:** $\mathcal{L}(\phi_1, \dot{\phi}_1, \dots) = T - W$.
- Step 8: Conjugate momenta:** $Q_n = \frac{\partial \mathcal{L}}{\partial \dot{\phi}_n}$ (these are the node charges).
- Step 9: Legendre transform:** $H = \sum_n Q_n \dot{\phi}_n - \mathcal{L}$.
- Step 10: Quantize:** $[\hat{\phi}_n, \hat{Q}_m] = i\hbar \delta_{nm}$.

Now let us see this in full detail with a worked example.

3. Worked Example: Two Nodes Coupled by a Capacitor

Consider two LC oscillators coupled by a capacitor C_3 . Node 1 has a capacitor C_1 to ground and an inductor L_1 to ground. Node 2 has a capacitor C_2 to ground and an inductor L_2 to ground. A coupling capacitor C_3 connects node 1 to node 2.



3.1 Step 1–4: Setup

Nodes: 0 (ground), 1, 2. Active nodes: $N = 2$.

Spanning tree: Choose the tree to include C_1 , C_2 , and C_3 (all capacitors). The inductors L_1 and L_2 become closure branches. (This is the preferred choice: placing capacitors in the tree simplifies the kinetic energy.)

Node fluxes: $\phi_1(t)$ and $\phi_2(t)$, defined as the time-integral of voltage from ground to each node along the tree.

3.2 Step 5–6: Energies in Terms of Node Fluxes

Kinetic Energy (Capacitive)

Each capacitor stores energy $\frac{1}{2}C_b\dot{\Phi}_b^2$, where $\dot{\Phi}_b$ is the voltage across it. From KVL:

- C_1 : connects node 1 to ground $\Rightarrow \dot{\Phi}_{C_1} = \dot{\phi}_1 - 0 = \dot{\phi}_1$
- C_2 : connects node 2 to ground $\Rightarrow \dot{\Phi}_{C_2} = \dot{\phi}_2$
- C_3 : connects node 1 to node 2 $\Rightarrow \dot{\Phi}_{C_3} = \dot{\phi}_1 - \dot{\phi}_2$

Therefore:

$$T = \frac{1}{2}C_1\dot{\phi}_1^2 + \frac{1}{2}C_2\dot{\phi}_2^2 + \frac{1}{2}C_3(\dot{\phi}_1 - \dot{\phi}_2)^2 \quad (5)$$

Expanding:

$$T = \frac{1}{2}(C_1 + C_3)\dot{\phi}_1^2 + \frac{1}{2}(C_2 + C_3)\dot{\phi}_2^2 - C_3\dot{\phi}_1\dot{\phi}_2 \quad (6)$$

This can be written compactly in matrix form:

$$T = \frac{1}{2}\dot{\boldsymbol{\phi}}^T \mathbf{C} \dot{\boldsymbol{\phi}}, \quad \mathbf{C} = \begin{pmatrix} C_1 + C_3 & -C_3 \\ -C_3 & C_2 + C_3 \end{pmatrix} \quad (7)$$

where $\dot{\boldsymbol{\phi}} = (\dot{\phi}_1, \dot{\phi}_2)^T$. The matrix $\mathbf{C}$ is the **capacitance matrix** of the network.

Potential Energy (Inductive)

Each inductor stores energy $\Phi_b^2/(2L_b)$. Since the inductors are closure branches connecting each node to ground:

- L_1 : connects node 1 to ground $\Rightarrow \Phi_{L_1} = \phi_1$ (no external flux assumed)
- L_2 : connects node 2 to ground $\Rightarrow \Phi_{L_2} = \phi_2$

$$W = \frac{\phi_1^2}{2L_1} + \frac{\phi_2^2}{2L_2} \quad (8)$$

3.3 Step 7: Lagrangian

$$\mathcal{L} = T - W = \frac{1}{2}(C_1 + C_3)\dot{\phi}_1^2 + \frac{1}{2}(C_2 + C_3)\dot{\phi}_2^2 - C_3\dot{\phi}_1\dot{\phi}_2 - \frac{\phi_1^2}{2L_1} - \frac{\phi_2^2}{2L_2} \quad (9)$$

3.4 Step 8: Conjugate Momenta (Node Charges)

Deriving the Node Charges

The conjugate momenta are the node charges:

$$Q_1 = \frac{\partial \mathcal{L}}{\partial \dot{\phi}_1} = (C_1 + C_3)\dot{\phi}_1 - C_3\dot{\phi}_2 = C_1\dot{\phi}_1 + C_3(\dot{\phi}_1 - \dot{\phi}_2) \quad (10)$$

$$Q_2 = \frac{\partial \mathcal{L}}{\partial \dot{\phi}_2} = (C_2 + C_3)\dot{\phi}_2 - C_3\dot{\phi}_1 = C_2\dot{\phi}_2 - C_3(\dot{\phi}_1 - \dot{\phi}_2) \quad (11)$$

In matrix form: $\vec{Q} = \mathbf{C} \dot{\vec{\phi}}$.

Physical interpretation: Q_1 is the *total charge* that has flowed into node 1 from ground. It equals the charge on C_1 plus the charge on C_3 (from the node-1 side): $Q_1 = C_1\dot{\phi}_1 + C_3(\dot{\phi}_1 - \dot{\phi}_2)$. The coupling capacitor C_3 causes Q_1 to depend on *both* $\dot{\phi}_1$ and $\dot{\phi}_2$ — this is the source of capacitive coupling in the Hamiltonian.

3.5 Step 9: Legendre Transform — The Hamiltonian

To perform the Legendre transform $H = \sum_n Q_n \dot{\phi}_n - \mathcal{L}$, we need to express $\dot{\phi}_n$ in terms of Q_n . This requires inverting the capacitance matrix.

Inverting the Capacitance Matrix

From $\vec{Q} = \mathbf{C} \dot{\vec{\phi}}$, we need $\dot{\vec{\phi}} = \mathbf{C}^{-1} \vec{Q}$.

The inverse of a 2×2 matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ is $\frac{1}{ad-bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$.

Define $D = \det \mathbf{C} = (C_1 + C_3)(C_2 + C_3) - C_3^2 = C_1C_2 + C_1C_3 + C_2C_3$.

Then:

$$\mathbf{C}^{-1} = \frac{1}{D} \begin{pmatrix} C_2 + C_3 & C_3 \\ C_3 & C_1 + C_3 \end{pmatrix} \quad (12)$$

So:

$$\dot{\phi}_1 = \frac{1}{D} [(C_2 + C_3)Q_1 + C_3Q_2] = \frac{1}{D} [C_2Q_1 + C_3(Q_1 + Q_2)] \quad (13)$$

$$\dot{\phi}_2 = \frac{1}{D} [C_3Q_1 + (C_1 + C_3)Q_2] = \frac{1}{D} [C_1Q_2 + C_3(Q_1 + Q_2)] \quad (14)$$

Now compute $H = \vec{Q}^T \dot{\vec{\phi}} - \mathcal{L}$. Since $T = \frac{1}{2} \dot{\vec{\phi}}^T \mathbf{C} \dot{\vec{\phi}} = \frac{1}{2} \vec{Q}^T \mathbf{C}^{-1} \vec{Q}$ and $\vec{Q}^T \dot{\vec{\phi}} = \vec{Q}^T \mathbf{C}^{-1} \vec{Q}$, we get:

$$H = \vec{Q}^T \mathbf{C}^{-1} \vec{Q} - \frac{1}{2} \vec{Q}^T \mathbf{C}^{-1} \vec{Q} + W = \frac{1}{2} \vec{Q}^T \mathbf{C}^{-1} \vec{Q} + W \quad (15)$$

Key Result

The Hamiltonian for the two-node coupled circuit is:

$$H = \frac{1}{2} \vec{Q}^T \mathbf{C}^{-1} \vec{Q} + \frac{\phi_1^2}{2L_1} + \frac{\phi_2^2}{2L_2} \quad (16)$$

Expanding the kinetic (charge) part:

$$H = \frac{(C_2 + C_3)Q_1^2 + 2C_3Q_1Q_2 + (C_1 + C_3)Q_2^2}{2D} + \frac{\phi_1^2}{2L_1} + \frac{\phi_2^2}{2L_2} \quad (17)$$

where $D = C_1C_2 + C_1C_3 + C_2C_3$.

The cross-term $\propto C_3Q_1Q_2$ in the Hamiltonian is the **capacitive coupling** between the two oscillators. If $C_3 = 0$, the two oscillators decouple into independent LC oscillators. The coupling arises because the charge on node 1 depends on the voltage at node 2 through the shared capacitor. In the qubit–resonator system of circuit QED, this is exactly how the qubit couples to the readout resonator — through a coupling capacitor.

After quantization, this term becomes $\propto \hat{Q}_1\hat{Q}_2 \propto (\hat{a}_1 - \hat{a}_1^\dagger)(\hat{a}_2 - \hat{a}_2^\dagger)$, which in the rotating-wave approximation reduces to the Jaynes–Cummings coupling $g(\hat{a}_1\hat{a}_2^\dagger + \hat{a}_1^\dagger\hat{a}_2)$.

3.6 Step 10: Quantization

Promote node fluxes and charges to operators:

$$[\hat{\phi}_n, \hat{Q}_m] = i\hbar \delta_{nm} \quad (18)$$

This is the canonical commutation relation (the sign convention here matches $[\hat{x}, \hat{p}] = i\hbar$ with $\phi \leftrightarrow x$ and $Q \leftrightarrow p$, noting that in the LC case one often writes $[\hat{\Phi}, \hat{Q}] = -i\hbar$ due to the fact that $\dot{Q} = +\partial H/\partial \Phi$ while $\dot{\Phi} = -\partial H/\partial Q$; the sign is a convention and does not affect the physics).

4. The Quantum Harmonic Oscillator — Full Derivation

4.1 From Hamiltonian to Ladder Operators

Starting from the quantized single-node Hamiltonian:

$$\hat{H} = \frac{\hat{\Phi}^2}{2L} + \frac{\hat{Q}^2}{2C} \quad (19)$$

with $[\hat{\Phi}, \hat{Q}] = -i\hbar$ and $\omega = 1/\sqrt{LC}$, $Z_c = \sqrt{L/C}$.

Constructing Creation/Annihilation Operators

We seek $\hat{a}, \hat{a}^\dagger$ such that $\hat{H} = \hbar\omega(\hat{a}^\dagger\hat{a} + \frac{1}{2})$ and $[\hat{a}, \hat{a}^\dagger] = 1$.

Sign subtlety. In the standard mechanical oscillator with $[\hat{x}, \hat{p}] = +i\hbar$, one defines $\hat{a} \propto (\hat{x} + i\hat{p}/m\omega)$. But our circuit convention is $[\hat{\Phi}, \hat{Q}] = -i\hbar$ (the opposite sign). This flips which linear combination is $\hat{a}$ vs. $\hat{a}^\dagger$. The correct definitions compatible with $[\hat{\Phi}, \hat{Q}] = -i\hbar$ are:

$$\hat{a} = \frac{1}{\sqrt{2\hbar Z_c}} (\hat{\Phi} - iZ_c\hat{Q}) \quad (20)$$

$$\hat{a}^\dagger = \frac{1}{\sqrt{2\hbar Z_c}} (\hat{\Phi} + iZ_c\hat{Q}) \quad (21)$$

Verification of $[\hat{a}, \hat{a}^\dagger] = 1$. Writing $\hat{a} = \alpha\hat{\Phi} + \beta\hat{Q}$ and $\hat{a}^\dagger = \alpha\hat{\Phi} + \gamma\hat{Q}$ with $\alpha = 1/\sqrt{2\hbar Z_c}$, $\beta = -iZ_c/\sqrt{2\hbar Z_c}$, $\gamma = +iZ_c/\sqrt{2\hbar Z_c}$:

$$\begin{aligned} [\hat{a}, \hat{a}^\dagger] &= \alpha\gamma [\hat{\Phi}, \hat{Q}] + \beta\alpha [\hat{Q}, \hat{\Phi}] \\ &= \alpha\gamma (-i\hbar) + \beta\alpha (+i\hbar) \\ &= \alpha (-i\hbar) (\gamma - \beta) = \frac{1}{\sqrt{2\hbar Z_c}} \cdot (-i\hbar) \cdot \frac{2iZ_c}{\sqrt{2\hbar Z_c}} \\ &= \frac{-i\hbar \cdot 2iZ_c}{2\hbar Z_c} = \frac{2\hbar Z_c}{2\hbar Z_c} = 1 \quad \checkmark \end{aligned} \quad (22)$$

(If you had used the opposite assignment — the “naive” one from the mechanical analogy without accounting for the sign of $[\hat{\Phi}, \hat{Q}]$ — you would get $[\hat{a}, \hat{a}^\dagger] = -1$. This is a common source of confusion.)

Let us settle the correct definitions and verify $\hat{H} = \hbar\omega(\hat{a}^\dagger\hat{a} + 1/2)$.

Key Result

With $[\hat{\Phi}, \hat{Q}] = -i\hbar$ and $Z_c = \sqrt{L/C}$, $\omega = 1/\sqrt{LC}$:

$$\hat{a}^\dagger = \frac{1}{\sqrt{2\hbar Z_c}} (\hat{\Phi} + iZ_c\hat{Q}), \quad \hat{a} = \frac{1}{\sqrt{2\hbar Z_c}} (\hat{\Phi} - iZ_c\hat{Q}) \quad (23)$$

$$\text{Inverting:} \quad \hat{\Phi} = \Phi_{\text{ZPF}} (\hat{a} + \hat{a}^\dagger), \quad \hat{Q} = -iQ_{\text{ZPF}} (\hat{a} - \hat{a}^\dagger) \quad (24)$$

where $\Phi_{\text{ZPF}} = \sqrt{\frac{\hbar Z_c}{2}} = \sqrt{\frac{L\hbar\omega}{2}}$ and $Q_{\text{ZPF}} = \sqrt{\frac{\hbar}{2Z_c}} = \sqrt{\frac{C\hbar\omega}{2}}$.

Verifying $\hat{H} = \hbar\omega(\hat{a}^\dagger\hat{a} + 1/2)$

Substitute $\hat{\Phi} = \Phi_{\text{ZPF}}(\hat{a} + \hat{a}^\dagger)$ and $\hat{Q} = -iQ_{\text{ZPF}}(\hat{a} - \hat{a}^\dagger)$ into $\hat{H}$:

Inductive term:

$$\frac{\hat{\Phi}^2}{2L} = \frac{\Phi_{\text{ZPF}}^2}{2L}(\hat{a} + \hat{a}^\dagger)^2 = \frac{\hbar Z_c}{4L}(\hat{a} + \hat{a}^\dagger)^2 = \frac{\hbar\omega}{4}(\hat{a} + \hat{a}^\dagger)^2 \quad (25)$$

using $\Phi_{\text{ZPF}}^2 = \hbar Z_c/2$ and $Z_c/(2L) = \omega/2$ (since $Z_c = \omega L$).

Capacitive term: Since $\hat{Q}^2 = (-iQ_{\text{ZPF}})^2(\hat{a} - \hat{a}^\dagger)^2 = -Q_{\text{ZPF}}^2(\hat{a} - \hat{a}^\dagger)^2$:

$$\frac{\hat{Q}^2}{2C} = \frac{-Q_{\text{ZPF}}^2}{2C}(\hat{a} - \hat{a}^\dagger)^2 = -\frac{\hbar}{4Z_c C}(\hat{a} - \hat{a}^\dagger)^2 = -\frac{\hbar\omega}{4}(\hat{a} - \hat{a}^\dagger)^2 \quad (26)$$

using $Q_{\text{ZPF}}^2 = \hbar/(2Z_c)$ and $1/(2Z_c C) = \omega/2$ (since $1/(Z_c C) = \omega$).

Combining:

$$\hat{H} = \frac{\hbar\omega}{4} \left[(\hat{a} + \hat{a}^\dagger)^2 - (\hat{a} - \hat{a}^\dagger)^2 \right] \quad (27)$$

Expand:

$$(\hat{a} + \hat{a}^\dagger)^2 = \hat{a}^2 + \hat{a}\hat{a}^\dagger + \hat{a}^\dagger\hat{a} + (\hat{a}^\dagger)^2 \quad (28)$$

$$(\hat{a} - \hat{a}^\dagger)^2 = \hat{a}^2 - \hat{a}\hat{a}^\dagger - \hat{a}^\dagger\hat{a} + (\hat{a}^\dagger)^2 \quad (29)$$

Subtracting: $(\hat{a} + \hat{a}^\dagger)^2 - (\hat{a} - \hat{a}^\dagger)^2 = 2(\hat{a}\hat{a}^\dagger + \hat{a}^\dagger\hat{a})$. Therefore:

$$\hat{H} = \frac{\hbar\omega}{2}(\hat{a}\hat{a}^\dagger + \hat{a}^\dagger\hat{a}) = \frac{\hbar\omega}{2}(2\hat{a}^\dagger\hat{a} + 1) = \hbar\omega \left(\hat{a}^\dagger\hat{a} + \frac{1}{2} \right) \quad \checkmark \quad (30)$$

where we used $\hat{a}\hat{a}^\dagger = \hat{a}^\dagger\hat{a} + [\hat{a}, \hat{a}^\dagger] = \hat{a}^\dagger\hat{a} + 1$.

5. Physical Realizations and the Qubit Problem

5.1 Lumped-Element and Transmission-Line Resonators

The quantum harmonic oscillator is physically realized as:

- **Lumped-element resonator**: A planar spiral inductor (providing L) integrated with an interdigitated or parallel-plate capacitor (providing C). Compact, with $\omega_r/2\pi \sim 4\text{--}8$ GHz.
- **Coplanar waveguide (CPW) resonator**: A superconducting strip (Nb on sapphire or Si) flanked by ground planes. Length $L \approx \lambda/2 \approx 25$ mm at 6 GHz. Coupling gaps at each end define the resonator boundaries.

The CPW resonator is a *distributed* system with per-unit-length capacitance c and inductance ℓ . It supports standing waves at frequencies $\omega_n = n\pi v/L$ where $v = 1/\sqrt{\ell c}$. Near each resonance, the distributed system maps to an effective LC oscillator.

For typical parameters ($\omega_r/2\pi \approx 6$ GHz, $C \sim 1$ pF, gap ~ 5 μm), the vacuum voltage fluctuations are:

$$V_{0,\text{rms}} = V_{\text{ZPF}} = \sqrt{\frac{\hbar\omega_r}{2C}} \approx 1 \text{ } \mu\text{V}, \quad E_{0,\text{rms}} \approx 0.2 \text{ V/m} \quad (31)$$

These fluctuations are real and measurable — they are responsible for the vacuum Rabi splitting observed in circuit QED experiments.

5.2 Why a Harmonic Oscillator Cannot Be a Qubit

The energy spectrum $E_n = \hbar\omega(n + 1/2)$ is **equally spaced**. This means:

1. The transition frequency $\omega_{n \rightarrow n+1} = \omega$ is the **same for all** n .
2. A resonant microwave photon ($E = \hbar\omega$) absorbed by the oscillator in $|0\rangle$ excites it to $|1\rangle$ — good.
3. But a second photon at the same frequency excites it to $|2\rangle$, then $|3\rangle$, *ad infinitum*. The population “leaks” to higher levels uncontrollably.
4. There is **no way** to selectively drive only the $|0\rangle \leftrightarrow |1\rangle$ transition because $\omega_{01} = \omega_{12} = \omega_{23} = \dots$

Conclusion: We need a **nonlinear, dissipationless** element that makes $\omega_{01} \neq \omega_{12}$. The **Josephson junction** is the only known circuit element with this property.

6. Phase and Flux Quantization

6.1 Flux Quantization in a Superconducting Loop

In a superconductor, all Cooper pairs share a single macroscopic wave function $\psi(\mathbf{r}) = \sqrt{n_s} e^{i\theta(\mathbf{r})}$. This wave function must be single-valued:

$$\psi(r, \theta) = \psi(r, \theta + 2\pi) \Rightarrow \Delta\theta|_{\text{around loop}} = 2\pi j, \quad j \in \mathbb{Z} \quad (32)$$

The **Aharonov-Bohm effect** says that a charge q moving around a closed loop acquires a phase:

$$\delta = \frac{q}{\hbar} \oint \vec{A} \cdot d\vec{s} = \frac{q}{\hbar} \Phi_{\text{enclosed}} \quad (33)$$

For a Cooper pair ($q = 2e$):

$$\delta = \frac{2e}{\hbar} \Phi \quad (34)$$

Single-valuedness $\delta = 2\pi j$ then gives:

Key Result

Flux quantization:

$$\Phi = j \phi_0, \quad \phi_0 = \frac{h}{2e} \approx 2.067 \times 10^{-15} \text{ Wb} \quad (35)$$

The magnetic flux through any closed superconducting loop is quantized in units of the **magnetic flux quantum** ϕ_0 .

The superconducting phase difference is related to flux by:

$$\delta = \frac{2\pi\Phi}{\phi_0} = \frac{\Phi}{\varphi_0} \quad (36)$$

where $\varphi_0 = \frac{h}{2e} = \frac{\phi_0}{2\pi}$ is the **reduced flux quantum**.

6.2 Phase-Number Commutator

Defining the Cooper pair number operator $\hat{N} = \hat{Q}/(2e)$ (the number of Cooper pairs that have tunneled) and the phase operator $\hat{\delta} = \hat{\Phi}/\varphi_0$, we can derive their commutator from $[\hat{\Phi}, \hat{Q}] = -i\hbar$:

Deriving $[\hat{\delta}, \hat{N}] = -i$

$$[\hat{\delta}, \hat{N}] = \left[\frac{\hat{\Phi}}{\varphi_0}, \frac{\hat{Q}}{2e} \right] = \frac{1}{\varphi_0 \cdot 2e} [\hat{\Phi}, \hat{Q}] = \frac{-i\hbar}{\frac{h}{2e} \cdot 2e} = \frac{-i\hbar}{\hbar} = -i \quad (37)$$

This is the **number-phase uncertainty relation**: $\Delta N \cdot \Delta\delta \geq \frac{1}{2}$.

If the phase is sharply defined ($\Delta\delta$ small), the number of Cooper pairs is highly uncertain, and vice versa. This complementarity underlies the entire zoo of superconducting qubit designs.

7. The Josephson Effect — Full Derivation

7.1 Physical Setup

A Josephson junction consists of two superconducting electrodes (S_1, S_2) separated by a thin insulating barrier (typically $\sim 1\text{--}2$ nm of AlO_x). Each electrode has a macroscopic wave function $\psi_i = \sqrt{n_i^{CP}} e^{i\delta_i}$ with a well-defined superconducting phase δ_i .

Key fact: The total number of Cooper pairs $N = N_1 + N_2$ is conserved, but Cooper pairs can tunnel through the barrier, changing N_1 and N_2 individually.

7.2 The Hilbert Space of the Junction

Define the state $|n\rangle$ as the state in which n Cooper pairs have tunneled from electrode 1 to electrode 2:

$$|n\rangle \equiv |N_1 - n, N_2 + n\rangle \quad (38)$$

The Hilbert space is spanned by $\{|n\rangle\}_{n=-\infty}^{+\infty}$, and any state is:

$$|\Psi\rangle = \sum_{n=-\infty}^{+\infty} c_n |n\rangle \quad (39)$$

7.3 The Josephson Hamiltonian (Tunneling Hamiltonian)

The tunneling process couples adjacent charge states: a Cooper pair tunneling from S_1 to S_2 takes $|n\rangle \rightarrow |n+1\rangle$, and vice versa. The tunneling matrix element is $E_J/2$:

Key Result

Josephson Hamiltonian:

$$\hat{H}_J = -\frac{E_J}{2} \sum_{n=-\infty}^{+\infty} \left[|n\rangle \langle n+1| + |n+1\rangle \langle n| \right] \quad (40)$$

with tunneling energy $E_J = \frac{1}{2} \left(\frac{\hbar}{2e} \right)^2 \frac{\Delta}{R_N}$, where Δ is the superconducting gap and R_N is the normal-state resistance of the junction. The parameter R_N is controlled by the junction area and barrier thickness.

This is mathematically a **tight-binding model** on a one-dimensional lattice of charge states — identical in structure to an electron hopping between sites in a crystal.

7.4 Phase States and the Cosine Potential

Diagonalizing the Josephson Hamiltonian

The eigenstates of Eq. (40) are found by Fourier-transforming from the “position” (number) basis to the “momentum” (phase) basis. Define the **phase state**:

$$|\delta\rangle = \sum_{n=-\infty}^{+\infty} e^{in\delta} |n\rangle \quad (41)$$

This is a coherent superposition over all charge states with well-defined phase δ but completely uncertain number (by the number-phase uncertainty). Note the periodicity: $|\delta + 2\pi\rangle = |\delta\rangle$.

Action of $\hat{H}_J$ on $|\delta\rangle$:

$$\begin{aligned}
 \hat{H}_J |\delta\rangle &= -\frac{E_J}{2} \sum_n \left[|n\rangle \langle n+1| + |n+1\rangle \langle n| \right] \sum_m e^{im\delta} |m\rangle \\
 &= -\frac{E_J}{2} \sum_n \left[e^{i(n+1)\delta} |n\rangle + e^{in\delta} |n+1\rangle \right] \\
 &= -\frac{E_J}{2} \sum_n \left[e^{i(n+1)\delta} |n\rangle + e^{i(n+1-1)\delta} |n+1\rangle \right]
 \end{aligned} \tag{42}$$

Re-index the second sum by $m = n + 1$:

$$\begin{aligned}
 \hat{H}_J |\delta\rangle &= -\frac{E_J}{2} \sum_n e^{i(n+1)\delta} |n\rangle - \frac{E_J}{2} \sum_m e^{i(m-1)\delta} |m\rangle \\
 &= -\frac{E_J}{2} \left(e^{i\delta} + e^{-i\delta} \right) \sum_n e^{in\delta} |n\rangle \\
 &= -E_J \cos \delta |\delta\rangle
 \end{aligned} \tag{43}$$

Therefore $|\delta\rangle$ is an eigenstate with eigenvalue $-E_J \cos \delta$, and we can write:

$$\boxed{\hat{H}_J = -E_J \cos \delta} \tag{44}$$

This is the **Josephson potential energy**: a cosine in the superconducting phase δ .

The tight-binding model $\rightarrow$ cosine dispersion is the *exact same mathematics* as Bloch theory in solid-state physics. The charge states $|n\rangle$ are “lattice sites,” E_J is the “hopping energy,” and δ is the “crystal momentum.” The cosine dispersion $E(\delta) = -E_J \cos \delta$ is the band structure. For a Josephson junction with additional charging energy ($E_C \neq 0$), the spectrum develops into Mathieu-function energy bands as a function of the offset charge — exactly like Bloch bands as a function of crystal momentum. In the transmon regime ($E_J/E_C \gg 1$), the bands become exponentially flat, meaning insensitivity to offset charge.

7.5 The First Josephson Relation — Full Derivation

Deriving $I = I_c \sin \delta$

The current through the junction is the rate of charge transfer. Since each tunneling event transfers a Cooper pair ($2e$), the current operator is:

$$\hat{I} = 2e \frac{d\hat{n}}{dt} = \frac{2e}{i\hbar} [\hat{n}, \hat{H}] \quad (45)$$

where $\hat{n} = \sum_n n |n\rangle \langle n|$ is the number operator.

Compute the commutator with $\hat{H}_J$. We need $[\hat{n}, |m\rangle \langle m+1|]$ and $[\hat{n}, |m+1\rangle \langle m|]$. Using $\hat{n}|m\rangle = m|m\rangle$:

$$\begin{aligned} [\hat{n}, |m\rangle \langle m+1|] &= \hat{n}|m\rangle \langle m+1| - |m\rangle \langle m+1| \hat{n} \\ &= m|m\rangle \langle m+1| - (m+1)|m\rangle \langle m+1| = -|m\rangle \langle m+1| \end{aligned} \quad (46)$$

Similarly: $[\hat{n}, |m+1\rangle \langle m|] = (m+1)|m+1\rangle \langle m| - m|m+1\rangle \langle m| = +|m+1\rangle \langle m|$.

Therefore:

$$[\hat{n}, \hat{H}_J] = -\frac{E_J}{2} \sum_m \left[-|m\rangle \langle m+1| + |m+1\rangle \langle m| \right] = \frac{E_J}{2} \sum_m \left[|m\rangle \langle m+1| - |m+1\rangle \langle m| \right] \quad (47)$$

Now evaluate on a phase state $|\delta\rangle$:

$$\begin{aligned} [\hat{n}, \hat{H}_J] |\delta\rangle &= \frac{E_J}{2} \sum_m \left[e^{i(m+1)\delta} |m\rangle - e^{im\delta} |m+1\rangle \right] \\ &= \frac{E_J}{2} \left(e^{i\delta} - e^{-i\delta} \right) \sum_m e^{im\delta} |m\rangle \\ &= \frac{E_J}{2} \cdot 2i \sin \delta |\delta\rangle = iE_J \sin \delta |\delta\rangle \end{aligned} \quad (48)$$

Therefore:

$$\hat{I} |\delta\rangle = \frac{2e}{i\hbar} \cdot iE_J \sin \delta |\delta\rangle = \frac{2eE_J}{\hbar} \sin \delta |\delta\rangle \quad (49)$$

Defining the critical current $I_c = \frac{2e}{\hbar} E_J = \frac{E_J}{\varphi_0}$:

$$I = I_c \sin \delta \quad (\text{1st Josephson relation}) \quad (50)$$

This says that a **dissipationless supercurrent** flows through the junction, with magnitude set by the sine of the phase difference — even at *zero applied voltage*. This is the **DC Josephson effect**.

7.6 The Second Josephson Relation — Full Derivation

Deriving $\dot{\delta} = V/\varphi_0$

Now apply a voltage V across the junction. The electrostatic energy for having n excess Cooper pairs on one electrode is $-2eVn$. In the semiclassical regime (well-defined V), the total Hamiltonian becomes:

$$\hat{H} = -E_J \cos \hat{\delta} - 2eV \hat{n} \quad (51)$$

The equation of motion for the phase operator:

$$\frac{d\hat{\delta}}{dt} = \frac{1}{i\hbar} [\hat{H}, \hat{\delta}] = \frac{1}{i\hbar} [-2eV\hat{n}, \hat{\delta}] = -\frac{2eV}{i\hbar} [\hat{n}, \hat{\delta}] \quad (52)$$

We need $[\hat{n}, \hat{\delta}]$. From $[\hat{\delta}, \hat{n}] = -i$ (derived in Section 6.2), we get $[\hat{n}, \hat{\delta}] = +i$. Therefore:

$$\frac{d\hat{\delta}}{dt} = -\frac{2eV}{i\hbar} \cdot i = \frac{2eV}{\hbar} = \frac{V}{\varphi_0} \quad (53)$$

$$\dot{\delta} = \frac{V}{\varphi_0} = \frac{2eV}{\hbar} \quad (\text{2nd Josephson relation}) \quad (54)$$

This says that the phase across the junction evolves at a rate proportional to the applied voltage.

AC Josephson effect: If $V = V_0$ (DC), then $\delta(t) = \delta_0 + (V_0/\varphi_0)t$, and the current oscillates:

$$I(t) = I_c \sin\left(\delta_0 + \frac{V_0}{\varphi_0}t\right) = I_c \sin\left(\delta_0 + \frac{2eV_0}{\hbar}t\right) \quad (55)$$

at frequency $f = V_0/\varphi_0$, with $1/\varphi_0 = 483.597891 \text{ MHz}/\mu\text{V}$. This oscillation frequency is used as a voltage standard.

Connection to electrodynamic flux: Integrating $\dot{\delta} = V/\varphi_0$:

$$\delta(t) = \frac{1}{\varphi_0} \int_{-\infty}^t V(t')dt' = \frac{\Phi(t)}{\varphi_0} \quad (56)$$

So the electrodynamic flux through the junction is $\Phi = \varphi_0 \delta$. This connects the abstract superconducting phase to the circuit variable flux.

8. The Josephson Inductance — Nonlinear Circuit Element

8.1 Derivation from the Josephson Relations

Deriving the Josephson Inductance $L_J(\delta)$

Starting from the classical Josephson relations:

$$I = I_c \sin \delta \quad (1\text{st JR})$$

$$V = \varphi_0 \dot{\delta} \quad (2\text{nd JR})$$

Differentiate the first relation:

$$\frac{dI}{dt} = I_c \cos \delta \dot{\delta} \quad (57)$$

Substitute $\dot{\delta} = V/\varphi_0$ from the second relation:

$$\frac{dI}{dt} = I_c \cos \delta \cdot \frac{V}{\varphi_0} \implies V = \frac{\varphi_0}{I_c \cos \delta} \frac{dI}{dt} \quad (58)$$

Comparing with the standard inductor relation $V = L \frac{dI}{dt}$:

$$L_J(\delta) = \frac{\varphi_0}{I_c \cos \delta} = \frac{\varphi_0}{I_c \sqrt{1 - (I/I_c)^2}} \quad (59)$$

8.2 Physical Interpretation

- At $\delta = 0$ (zero current): $L_J(0) = \varphi_0/I_c = \hbar/(2eI_c)$, a constant value.
- As $\delta \rightarrow \pi/2$ (current approaching I_c): $L_J \rightarrow \infty$ — the junction becomes infinitely “soft.”
- The inductance is **current-dependent**: it increases as more current flows. This is the *non-linearity*.

8.3 From Quadratic to Cosine Potential

For a **linear inductor**, the potential energy is parabolic:

$$U_L = \frac{\Phi^2}{2L} \longleftrightarrow \text{equally spaced energy levels} \quad (60)$$

For a **Josephson junction**, the potential energy is the cosine:

$$U_J = -E_J \cos\left(\frac{\Phi}{\varphi_0}\right) = E_J \left(1 - \cos\left(\frac{\Phi}{\varphi_0}\right)\right) \quad (\text{shifted so that minimum is at 0}) \quad (61)$$

Taylor expansion: recovering the linear inductor

For small phase ($|\Phi/\varphi_0| \ll 1$), expand the cosine:

$$\begin{aligned} E_J \left(1 - \cos\left(\frac{\Phi}{\varphi_0}\right)\right) &\approx E_J \left(\frac{1}{2} \frac{\Phi^2}{\varphi_0^2} - \frac{1}{24} \frac{\Phi^4}{\varphi_0^4} + \dots\right) \\ &= \frac{\Phi^2}{2L_J(0)} - \frac{E_J}{24} \frac{\Phi^4}{\varphi_0^4} + \dots \end{aligned} \quad (62)$$

where $L_J(0) = \varphi_0^2/E_J = \varphi_0/I_c$.

The first term is a harmonic oscillator with inductance $L_J(0)$. The second term ($\propto \Phi^4$) is

the **leading nonlinear correction**. It is this quartic term that creates the anharmonicity needed for qubit operation.

Since the coefficient of Φ^4 is *negative*, the cosine potential is “softer” than the parabola at large amplitudes. This means:

- Higher energy levels are spaced *less* than $\hbar\omega$.
- The anharmonicity $\alpha = \omega_{12} - \omega_{01} < 0$ (negative anharmonicity).
- This is characteristic of the **transmon qubit**.

Conversely, for a “hardening” nonlinearity (Φ^4 with positive coefficient), one would get *positive anharmonicity*.

9. The Full Junction Hamiltonian and the Transmon

9.1 The Junction as a Parallel L_J – C Circuit

A real Josephson junction has a self-capacitance C (the two electrodes separated by the oxide form a parallel-plate capacitor). Thus the junction is modeled as a nonlinear inductor (L_J) in parallel with a capacitor (C).

Applying the circuit quantization recipe to this single-node circuit:

The Josephson Junction Hamiltonian

Kinetic energy (capacitive): $T = \frac{1}{2}C\dot{\Phi}^2$

Potential energy (Josephson): $U = E_J(1 - \cos(\Phi/\varphi_0))$

Lagrangian: $\mathcal{L} = \frac{1}{2}C\dot{\Phi}^2 - E_J(1 - \cos(\Phi/\varphi_0))$

Conjugate momentum: $Q = \partial\mathcal{L}/\partial\dot{\Phi} = C\dot{\Phi}$

Hamiltonian:

$$H = Q\dot{\Phi} - \mathcal{L} = \frac{Q^2}{2C} + E_J \left(1 - \cos \frac{\Phi}{\varphi_0}\right) \quad (63)$$

Switching to number and phase variables ($\delta = \Phi/\varphi_0$, $Q = 2en$):

$$H = 4E_C \hat{n}^2 - E_J \cos \hat{\delta} \quad (64)$$

where $E_C = e^2/(2C)$ is the single-electron charging energy (so $4E_C$ is the Cooper-pair charging energy $(2e)^2/(2C)$). We dropped the constant E_J .

9.2 The Two Competing Energy Scales

The Hamiltonian (64) has two terms with opposing tendencies:

The E_J/E_C Competition

1. **Charging energy** $4E_C \hat{n}^2$: penalizes charge fluctuations. Favors states with well-defined n (localized in number space).
2. **Josephson energy** $-E_J \cos \hat{\delta}$: favors tunneling (delocalization in number space, localization in phase space).

The number-phase uncertainty $\Delta n \cdot \Delta \delta \geq 1/2$ means these tendencies are **mutually exclusive**. The ratio E_J/E_C is the single most important design parameter.

9.3 Limiting Regimes

Charge regime ($E_J/E_C \ll 1$): Charging energy dominates. The eigenstates are approximately charge states $|n\rangle$. The system is highly sensitive to gate charge (“offset charge”) n_g : the Hamiltonian becomes $4E_C(\hat{n} - n_g)^2 - E_J \cos \hat{\delta}$, and the transition frequency depends strongly on n_g , which fluctuates uncontrollably due to charge noise. This is the original Cooper pair box (Bouchiat et al., 1998).

Transmon regime ($E_J/E_C \gg 1$, typically $E_J/E_C \sim 50$ – 100): Josephson energy dominates. The phase δ is approximately well-defined, and the “particle” oscillates near the bottom of the cosine well. The charge sensitivity is **exponentially suppressed** as $\sim e^{-\sqrt{8E_J/E_C}}$ (Koch et al., PRA 2007). The price: **reduced anharmonicity** $|\alpha|/\omega_q \sim (E_C/\hbar\omega_q)$.

9.4 Transmon Spectrum — Deriving ω_q and α

Perturbative treatment in the transmon regime

In the limit $E_J/E_C \gg 1$, expand the cosine to 4th order around $\delta = 0$:

$$E_J(1 - \cos \delta) \approx \frac{E_J}{2} \delta^2 - \frac{E_J}{24} \delta^4 \quad (65)$$

The quadratic part gives a harmonic oscillator with:

$$\omega_p = \frac{1}{\sqrt{L_J(0)C}} = \frac{1}{\varphi_0} \sqrt{\frac{E_J}{C}} = \frac{\sqrt{8E_J E_C}}{\hbar} \quad (66)$$

(this is called the **plasma frequency** of the junction). We can write $\delta = (\delta_{\text{ZPF}})(\hat{a} + \hat{a}^\dagger)$ where $\delta_{\text{ZPF}} = (2E_C/E_J)^{1/4}$ is the zero-point fluctuation of the phase.

The quartic correction $-\frac{E_J}{24} \delta^4$ is treated as a perturbation. In terms of ladder operators:

$$\delta^4 = \delta_{\text{ZPF}}^4 (\hat{a} + \hat{a}^\dagger)^4 \quad (67)$$

The expectation value of $(\hat{a} + \hat{a}^\dagger)^4$ in the number state $|n\rangle$ (keeping only the terms that survive normal ordering and contribute to the diagonal $\langle n | \dots | n \rangle$):

$$\langle n | (\hat{a} + \hat{a}^\dagger)^4 | n \rangle = 6n^2 + 6n + 3 \quad (68)$$

The perturbative energy shift of level n is:

$$\begin{aligned} \Delta E_n &= -\frac{E_J}{24} \delta_{\text{ZPF}}^4 (6n^2 + 6n + 3) \\ &= -\frac{E_J}{24} \left(\frac{2E_C}{E_J} \right) (6n^2 + 6n + 3) \\ &= -\frac{E_C}{12} (6n^2 + 6n + 3) = -\frac{E_C}{2} \left(n^2 + n + \frac{1}{2} \right) \end{aligned} \quad (69)$$

where we used $\delta_{\text{ZPF}}^4 = (2E_C/E_J)$.

The total energy is:

$$E_n \approx -E_J + \sqrt{8E_J E_C} \left(n + \frac{1}{2} \right) - \frac{E_C}{2} \left(n^2 + n + \frac{1}{2} \right) \quad (70)$$

The transition frequencies are:

$$\hbar \omega_{01} = E_1 - E_0 = \sqrt{8E_J E_C} - E_C \quad (71)$$

$$\hbar \omega_{12} = E_2 - E_1 = \sqrt{8E_J E_C} - 2E_C \quad (72)$$

The **anharmonicity**:

$$\alpha = \omega_{12} - \omega_{01} = -\frac{E_C}{\hbar} \quad (73)$$

And the **qubit frequency**:

$$\omega_q \equiv \omega_{01} = \frac{\sqrt{8E_J E_C} - E_C}{\hbar} \approx \frac{\sqrt{8E_J E_C}}{\hbar} \quad \text{for } E_J/E_C \gg 1 \quad (74)$$

Key Result

The transmon Hamiltonian, written in terms of ladder operators:

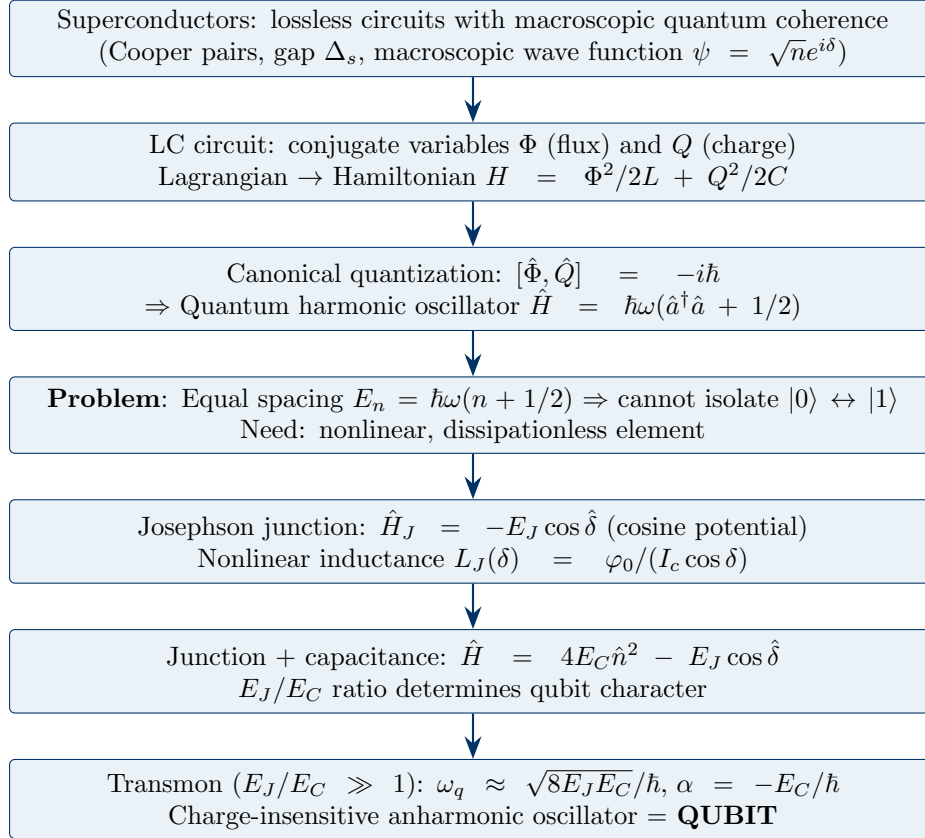
$$\hat{H} = \hbar \left[\frac{\omega_q}{2} + \left(\omega_q + \alpha \hat{a}^\dagger \hat{a} \right) \hat{a}^\dagger \hat{a} \right] \quad (75)$$

The key feature: the $\alpha (\hat{a}^\dagger \hat{a})^2 = \alpha \hat{n}(\hat{n} - 1)/\dots$ term makes the transition frequency **excitation-number-dependent**:

$$\omega_{n,n+1} = \omega_q + \alpha \cdot n \quad (76)$$

Since $\alpha < 0$, higher transitions are at *lower* frequency. For typical values $\omega_q/2\pi \approx 5$ GHz and $|\alpha|/2\pi \approx 200$ –300 MHz, the qubit transition ω_{01} is well-separated from ω_{12} , allowing selective addressing.

10. Synthesis: The Complete Logical Chain



10.1 What Makes This Work

The entire edifice rests on three physical pillars:

1. **Superconductivity** provides dissipationless circuit elements (no resistance $\Rightarrow$ no decoherence from ohmic losses) and a macroscopic quantum ground state separated from excitations by a gap Δ_s .
2. **The Josephson junction** provides the only known nonlinear, dissipationless circuit element. Its $\cos \delta$ potential breaks the harmonic equal-spacing degeneracy while introducing no loss.
3. **Millikelvin temperatures** ($T \sim 15$ mK) ensure $k_B T \ll \hbar\omega_q$ so that thermal excitations are negligible ($\bar{n}_{\text{th}} \sim 10^{-3}$ at 5 GHz).

10.2 Key Numbers for a Typical Transmon

Parameter	Typical Value	Formula
Qubit frequency $\omega_q/2\pi$	4–6 GHz	$\approx \sqrt{8E_J E_C}/\hbar$
Anharmonicity $ \alpha /2\pi$	200–300 MHz	$E_C/\hbar$
$E_J/\hbar$	10–30 GHz	$\varphi_0 I_c$
$E_C/\hbar$	200–300 MHz	$e^2/(2Ch)$
E_J/E_C	30–100	
Junction capacitance C	50–80 fF	
Critical current I_c	20–50 nA	
Josephson inductance $L_J(0)$	5–15 nH	φ_0/I_c
Operating temperature	10–20 mK	

11. Complete Formula Reference

11.1 Fundamental Constants

$$\phi_0 = \frac{h}{2e} \approx 2.067 \times 10^{-15} \text{ Wb} \quad (\text{magnetic flux quantum}) \quad (77)$$

$$\varphi_0 = \frac{\hbar}{2e} = \frac{\phi_0}{2\pi} \approx 3.291 \times 10^{-16} \text{ Wb} \quad (\text{reduced flux quantum}) \quad (78)$$

$$R_Q = \frac{h}{(2e)^2} \approx 6453.2 \, \Omega \quad (\text{resistance quantum for Cooper pairs}) \quad (79)$$

11.2 Harmonic Oscillator (LC Circuit)

$$\omega = 1/\sqrt{LC}, \quad Z_c = \sqrt{L/C} \quad (80)$$

$$\hat{H} = \frac{\hat{\Phi}^2}{2L} + \frac{\hat{Q}^2}{2C} = \hbar\omega \left(\hat{a}^\dagger \hat{a} + \frac{1}{2} \right) \quad (81)$$

$$\hat{\Phi} = \Phi_{\text{ZPF}}(\hat{a} + \hat{a}^\dagger), \quad \hat{Q} = -iQ_{\text{ZPF}}(\hat{a} - \hat{a}^\dagger) \quad (82)$$

$$\Phi_{\text{ZPF}} = \sqrt{\frac{\hbar Z_c}{2}}, \quad Q_{\text{ZPF}} = \sqrt{\frac{\hbar}{2Z_c}} \quad (83)$$

$$V_{\text{ZPF}} = \omega \Phi_{\text{ZPF}} = \sqrt{\frac{\hbar\omega}{2C}}, \quad \Phi_{\text{ZPF}} \cdot Q_{\text{ZPF}} = \frac{\hbar}{2} \quad (84)$$

11.3 Josephson Junction

$$\text{1st Josephson relation: } I = I_c \sin \delta \quad (85)$$

$$\text{2nd Josephson relation: } \dot{\delta} = V/\varphi_0 = 2eV/\hbar \quad (86)$$

$$\text{Josephson energy: } E_J = \varphi_0 I_c = \frac{\hbar I_c}{2e} \quad (87)$$

$$\text{Josephson inductance: } L_J(\delta) = \frac{\varphi_0}{I_c \cos \delta} = \frac{\varphi_0^2}{E_J \cos \delta} \quad (88)$$

$$\text{Josephson potential: } U_J = E_J(1 - \cos \delta) \quad (89)$$

$$\text{Charging energy: } E_C = \frac{e^2}{2C} \quad (90)$$

11.4 Transmon Qubit

$$\hat{H} = 4E_C \hat{n}^2 - E_J \cos \hat{\delta} \quad (91)$$

$$\omega_q \approx \frac{\sqrt{8E_J E_C}}{\hbar} - \frac{E_C}{\hbar} \quad (92)$$

$$\alpha = -\frac{E_C}{\hbar} \quad (\text{anharmonicity}) \quad (93)$$

$$E_n \approx -E_J + \sqrt{8E_J E_C} \left(n + \frac{1}{2} \right) - \frac{E_C}{2} \left(n^2 + n + \frac{1}{2} \right) \quad (94)$$

$$\omega_{n,n+1} = \omega_q + n\alpha \quad (95)$$

References: Devoret, Les Houches (1995); Vool & Devoret, arXiv:1610.03438 (2016); Koch et al., PRA 76, 042319 (2007); Burkard, Koch & DiVincenzo, PRB 69 (2004); Bishop thesis, Yale (2010).